



POLITECNICO
MILANO 1863

High Performance Scientific Computing for Aerospace Project

11th November 2025 update

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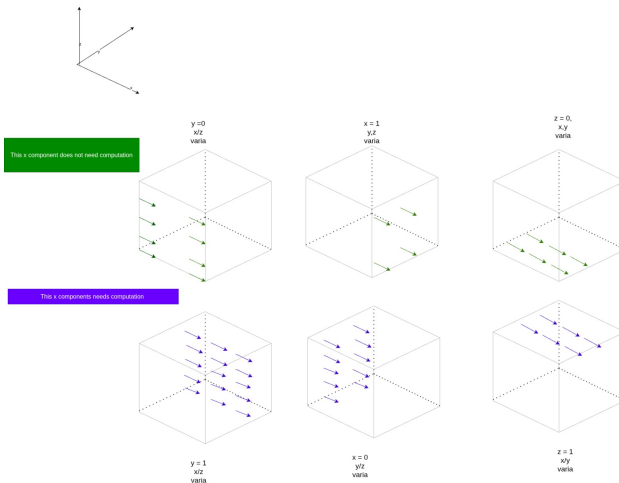
Content for slide 1 goes here.

Structure of the input file:

- **Mesh points:** Number of points in X, Y, Z directions
 - ▶ Example: 50 50 50 $\Rightarrow N_x = 50, N_y = 50, N_z = 50$
- **Time parameters:**
 - ▶ dt - Time step size, e.g., 0.01
 - ▶ T - Total simulation time, e.g., 10.0
- **Spatial step sizes:**
 - ▶ dx dy dz - Grid spacing in X, Y, Z, e.g., 0.5 0.5 0.5
- **Initial conditions files:**
 - ▶ u0_init_file - File with initial velocity values
 - ▶ p0_init_file - File with initial pressure values
 - ▶ k_file - File with permeability values

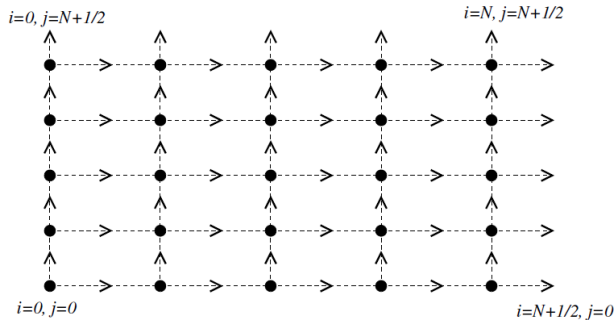
Boundary conditions implementation

3/7



Boundary conditions implementation pt.2

4/7



Code Validation

- Compute error
- Compute residual
- Convergence study

Parallelization Strategy

- Vectorization of the sequential code
- Parallelization of the code

Content for slide 5 goes here.

Content for slide 6 goes here.